

Practical #10

Practical Title:

To implement a basic CI pipeline by integrating GitHub, automated testing, and Jenkins jobs

Theory:

1. What is Continuous Integration (CI)?

Continuous Integration (CI) is a DevOps practice where developers frequently push code to a shared repository, and each change is automatically built and tested.

👉 Goal: Detect errors early and improve software quality.

2. Why is GitHub used in a CI pipeline?

GitHub is used because:

- Stores source code centrally
 - Tracks version history
 - Triggers CI pipelines (via webhook)
 - Enables team collaboration
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3. Role of Automated Testing in CI

Automated testing ensures that:

- Code changes do not break existing functionality
- Bugs are detected early
- Testing is fast and repeatable

👉 Tools like Selenium are commonly used.

4. Role of Jenkins in CI

Jenkins automates the CI pipeline by:

- Fetching code from GitHub
- Running build and test scripts
- Providing build status (success/failure)
- Triggering pipelines automatically

Procedure:

Step 1: Create or use an existing GitHub repository

Step 2: Add project files and test scripts to the repository

Step 3: Commit and push the code to GitHub

Step 4: Install and open Jenkins dashboard

Step 5: Create a new Jenkins job or pipeline

Step 6: Connect Jenkins with GitHub repository

Step 7: Configure build steps and test execution

Step 8: Run the Jenkins job

Step 9: Verify build status and console output

Step 10: Make changes in GitHub and trigger the pipeline again

Output Screenshots:

The screenshot shows the Jenkins web interface with the console output for a build named 'student-feedback-test'. The left sidebar shows the build's status as 'Success'. The console output displays the following information:

```
Started by user admin
Running as SYSTEM
Building as workspace C:\ProgramData\Jenkins\workspace\student-feedback-test
The recommended git sver is: NONE
No credentials opastier
> git ase reorgares --restive git dir C:\ProgramData\Jenkins\workspace\student-feedback-test.git timeout=10
Fetching changes from the remote Git reestlory
> git.ose ondng/remotes/rigin ord https://github.com/sharvayuzade/studentt-feedback.git # timeout=10
Fetching systroae changes from https://github.com/
> git.oas --version 2.41.1.44.01.1.
> git --version 8" git version 2.61.4.8.0.1.windows.1
> git.ose fetch --refs/remotes/origin/no2aa331 -- https://github.com/sharvayuzade/stud-feedback.git (refs/remotes/.origin/main/main)* # timeout=10
> git.oas mun-verse rrefs/remotes/9046667a066990410e4c05f6.34
Checking out fecthon {f206599407506567006340030e00000000635a190906630d6343d68221.b0t} # timeout=10
> git.ssa reotlp,uiso-aperveniaation # timeout=11 timesut=10
> git.ome chadback -1 ff2ecf908075d66e7a0a6904670e-064c056fc34
C:\ProgramData\Jenkins\workspace\student-feedback-test> cd "C:\Users\MDUMIA\MDUMIA\DevOps_website\Student-FedBack\student.feedback
C:\Users\MDUMIA\MDUMIA\DevOps_website\Student-feedback-test> dir JEBB08.NON\site-packages (4.1.30)
Requirement already satisfied: Secsfl.10 c:\python310\site-packages (from urilly (1.26.23))
Requirement already satisfied: certifi aAM01.2.19 in c:\python310\site-packages (from ulibs (1.26.15))
Requirement already satisfied: loba-2.2.0 (from certifi 8AM01.2.19) in c:\python310\site-packages (from certifi 0.20.23)
Requirement already satisfied: iniansaan-0.0.0 0.02.0.08 in c:\python310\site-packages (from satiod) (1.02.6)
Requirement already satisfied: abhon.1.8.1s 0.0.0.20 in c:\python310\site-packages (from scalin 1.5.0 (0.22.9))
Requirement already satisfied: atho22-ps-4.4.1 ljn c:\python310\site-packages (from actveni.2.0-1.4.8.91.0 (1.22.b)
Requirement already satisfied: sepo-adox-1.2.4.5 in c:\python310\site-packages (from trenovs1.13.6, 1.66.0 (1.200.6)
Requirement already satisfied: wecnous-0.03.0 in c:\python310\site-packages (from venopect-0.6503-1.0.a-12-3.ede-tam (1.45.1.286)
Requirement already satisfied: melensam.0.1.0.94 in c:\python310\site-packages (from troam-narsbox.2.3.4.a) (1.106.0)
Requirement already satisfied: tero09-4.0.2.1 haM01 in c:\python310\site-packages (from trevo-1.9.0 (1.232.8))

Git Build Data

Revision: ff2ecf908075d66e7a0a6904670e-064c056fc34
Repository: https://github.com/sharvayuzade/studindtent-feedback.git

Built Branches

• refs/remotes/origin: ff2ecf908075d66e7a0a6904670e064c056fc334. (nef/remutx/origin/main)(refs/remotes/origin/main)
```

Activity / Task:

Task 1: Jenkins Job

Repository Link: https://github.com/sharvayuzade/DevOps_CA2

Task 2: Build Result

- **Successful Build:**
 - Status: SUCCESS
 - Tests executed without errors
 - Console shows successful execution
- **Failed Build:**
 - Status: FAILURE
 - Errors in test or code

- Console shows error logs
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Task 3: CI Pipeline Flow

Complete Flow:

1. Developer writes code
 2. Pushes code to GitHub
 3. GitHub triggers Jenkins (webhook/polling)
 4. Jenkins fetches latest code
 5. Jenkins builds project
 6. Automated tests run
 7. Jenkins shows result (pass/fail)
 8. Feedback sent to developer
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Advantages

1. Early bug detection
 2. Faster development cycle
 3. Automated testing
 4. Improved code quality
 5. Continuous feedback
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Challenges

1. Initial setup complexity
2. Requires proper test scripts
3. Maintenance of pipeline